



Biology 2 Performance Targets

for the
South Carolina College- and Career-Ready Science Standards 2021

For use 2026-2027

July 2026

Ellen E. Weaver

State Superintendent of Education

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Purpose and Use

Science is a way of understanding the physical universe using observation and experimentation to explain natural phenomena. Science also refers to an organized body of knowledge that includes core ideas to the disciplines and common themes that bridge the disciplines. As science educators we must take a 3-dimensional approach in facilitating student learning. By addressing content, science and engineering practices and crosscutting concepts, students can have relevant and evidence-based instruction that can help solve current and future problems.

This document is intended as a guide for discerning and describing features of students and their work who have met the stated Performance Expectation (PE). This is not a curriculum or a means to limit instruction in the classroom. Although each PE states a dedicated Science and Engineering Practice (SEP) and Crosscutting Concept (CCC), students will need to use the whole range of SEPs and CCCs to achieve success by the end of instruction.

The PEs aligned to the course Biology 2 include four repeated from Biology 1. The expectation is that, through their Biology 2 coursework, students build upon the foundational knowledge developed in Biology 1 to deepen their understanding of the content. Thus, the Performance Target description of student performance may extend beyond the State Assessment Boundary for the four repeated PEs. Teachers have the flexibility to enrich students learning as best fits the needs and interests of their student population. Biology 2 is not assessed by the End of Course Examination Program. The PEs aligned to Biology 2 are:

- | | | |
|--------------------|--------------------|--------------------|
| • B-LS1-1 (repeat) | • B-LS2-6 | • B-LS3-3 (repeat) |
| • B-LS2-2 | • B-LS2-8 | • B-LS4-1 (repeat) |
| • B-LS2-3 | • B-LS3-1 | • B-LS4-3 |
| • B-LS2-4 | • B-LS3-2 (repeat) | • B-LS4-6 |

Three-dimensional science learning requires discipline specific communication skills. This means that effective science learning occurs when students are expected to speak, listen, read, and write in ways that are appropriate to science. With each Performance Target, there are question/sentence stems and terminology to support student discourse about phenomena to help teachers facilitate the acquisition of science discourse. Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding. The terms and stems in this section are intended to provide a baseline for teachers, neither list is exhaustive nor complete.

In addition to the doing (SEP), thinking (CCC), and learning of science knowledge (Disciplinary Core Ideas) outlined here, students will also require a working knowledge of grade-level appropriate tools and techniques of science. Students should know and recognize how scientists and engineers use these tools and techniques, not just identify them. Students should be able to use these tools to gather data, describe how these tools gather data, and/or interpret data sampled from them.

Document Updates

July 2026

- Addition of Appendix A: Biology 2 Vertical Articulation of Disciplinary Core Idea Instructional Focus Statements
- Adjusted formatting and grammar.

July 2025

- All Performance Expectation statements have been reformatted to call out each of the dimensions as follows:
 - Science and Engineering Practice – **bold**
 - Crosscutting Concept – *italicize*
 - Disciplinary Core Idea – regular
- The watermark from previous versions of this resource has been replaced with the wording “For use 2025-2026” on the title page and in the footer. This change was made to improve accessibility of this resource.
- Because scientific notation is no longer an expectation of the math standards, the following PEs have a new statement related to the use and application of scientific notation.
 - B-LS1-1
 - B-LS2-2
 - B-LS2-6
 - B-LS3-1
 - B-LS4-1
 - B-LS4-3
- B-LS1-1
 - additional terminology: peptide
- B-LS2-4
 - additional terminology: sugar

June 2024

- Updated watermark to 2024-2025.
- Adjusted formatting and grammar.

LS1 – From Molecules to Organisms: Structures and Processes

B-LS1-1. Construct an explanation based on evidence for how the structure of DNA determines *the structure of proteins, which carry out the essential functions* of life through systems of specialized cells.

State Assessment Boundary: Assessment does not include identification of specific cell or tissue types, whole body systems, specific protein structures and functions, or the biochemistry of protein synthesis.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Constructing Explanations and Designing Solutions Constructing explanations and designing solutions in 9-12 builds on K-8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific ideas, principles, and theories. Construct an explanation based on valid and reliable evidence obtained from a variety of sources (including [models, peer review, simulations, theories, students' own investigations]) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future.	LS1.A: Structure and Function Systems of specialized cells within organisms help them perform the essential functions of life. All cells contain genetic information, in the form of DNA. Genes are specific regions within the extremely large DNA molecules that form the chromosomes. Genes contain the instructions that code for the formation of molecules called proteins, which carry out most of the work of cells to perform the essential functions of life. Proteins provide structural components, serve as signaling devices, regulate cell activities, and determine the performance of cells through their enzymatic actions LS3.A: Inheritance of Traits The sequence of nucleotides spells out the information in a gene. DNA controls the expression of proteins by being transcribed into a “messenger” RNA, which is translated in turn by the cellular machinery into a protein.	Structure and Function Investigating or designing new systems or structures requires a detailed examination of the properties of different materials, the structures of different components, and connections of components to reveal its function.

*Because DNA molecules contain billions of base pairs, students may need to understand and apply the conventions of scientific notation when working with quantities of measurement and their calculations.

Observable features of student performance by the end of the course:

1. Articulating the explanations of phenomena

- a. Students articulate a statement describing/explaining relationships between the structure of DNA and the structure and function of resulting molecules, including:
 - i. regions of DNA determine the structure of proteins or functional RNA molecules and
 - ii. proteins carry out the essential functions of life.

2. Evidence

- a. Students identify and describe the evidence to construct the explanation, including:
 - i. all cells contain DNA,
 - ii. DNA contains regions called genes,
 - iii. a single gene codes for a protein or functional RNA molecules,
 - iv. through the biochemical process of transcription (initiation, elongation, termination), a DNA sequence is transcribed into a mRNA sequence,
 - v. through the biochemical process of translation (initiation, elongation, termination), a mRNA sequence is translated into an amino acid sequence at the ribosome,
 - vi. the sequence of genes contains instructions that code for proteins or functional RNA molecules,
 - vii. proteins carry out the essential functions of life, and
 - viii. in complex organisms, groups of specialized cells use proteins to carry out functions that are essential to the organism.

3. Reasoning

- a. Students use the following chain of reasoning to connect the evidence and support, refute, or revise an explanation, including:
 - i. because all cells contain DNA, all cells contain genes that can code for the formation of proteins and functional RNA molecules,
 - ii. gene sequence, including mutations, affects product function,
 - iii. proper functioning of many gene products is necessary for the proper functioning of cells,
 - iv. because of differentiation, different genes are activated in different cells that share the same genetic code in complex organisms, and
 - v. systems of specialized cells share similar structures and functions; these are carried out by proteins and functional RNA molecules.

B-LS1-1 Academic Language

Question/Sentence Stems

- The _____ structures help _____ to function because _____.
- The _____ structures are present in _____ and are related to the function _____.
- If ____ structure is altered, what results?
- What are some similarities and differences between ____ and ____?

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | |
|--|--------------------------|
| • Adenine (A) | • mutagen |
| • amino acid | • mutation |
| • anticodon | • nuclear membrane |
| • base | • nucleic acid |
| • cell | • nucleotide |
| • chromosome | • nucleus |
| • codon | • peptide |
| • cytoplasm | • peptide bond |
| • Cytosine (C) | • phosphate |
| • deoxyribose | • polypeptide |
| • differentiation | • protein synthesis |
| • deoxyribonucleic acid (DNA) | • ribose |
| • double helix | • ribosome |
| • elongation | • ribonucleic acid (RNA) |
| • endoplasmic reticulum (smooth and rough) | • ribosomal RNA (rRNA) |
| • enzyme | • start codon |
| • functional RNA | • stop codon |
| • gene | • termination |
| • Golgi apparatus | • Thymine (T) |
| • guanine | • transcription |
| • hydrogen bond | • translation |
| • initiation | • transfer RNA (tRNA) |
| • messenger RNA (mRNA) | • Uracil (U) |
| | • vesicle |

LS2 – Ecosystems: Interactions, Energy, and Dynamics

B-LS2-2. Use mathematical representations to support and revise explanations based on evidence about factors affecting biodiversity and populations in ecosystems of *different scales*.

Clarification Statement: Examples of mathematical representations include finding the average, determining trends, and using graphical comparisons of multiple sets of data.

State Assessment Boundary: Assessment is limited to provided data.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Using Mathematics and Computational Thinking Mathematical and computational thinking in 9-12 builds on K-8 experiences and progresses to using algebraic thinking and analysis, a range of linear and nonlinear functions including trigonometric functions, exponentials and logarithms, and computational tools for statistical analysis to analyze, represent, and model data. Simple computational simulations are created and used based on mathematical models of basic assumptions. Use mathematical representations of phenomena or design solutions to support and revise explanations.	LS2.A: Interdependent Relationships in Ecosystems Ecosystems have carrying capacities, which are limits to the numbers of organisms and populations they can support. These limits result from such factors as the availability of living and nonliving resources and from such challenges such as predation, competition, and disease. Organisms would have the capacity to produce populations of great size were it not for the fact that environments and resources are finite. This fundamental tension affects the abundance (number of individuals) of species in any given ecosystem. LS2.C: Ecosystem Dynamics, Functioning, and Resilience A complex set of interactions within an ecosystem can keep its numbers and types of organisms relatively constant over long periods of time under stable conditions. If a modest biological or physical disturbance to an ecosystem occurs, it may return to its more or less original status (i.e., the ecosystem is resilient), as opposed to becoming a very different ecosystem. Extreme fluctuations in conditions or the size of any population, however, can challenge the functioning of ecosystems in terms of resources and habitat availability.	Scale, Proportion, and Quantity Using the concept of orders of magnitude allows one to understand how a model at one scale relates to a model at another scale.

*Because populations can be large, students may need to understand and apply the conventions of scientific notation when working with quantities of measurement and their calculations.

Observable features of student performance by the end of the course:

1. Representation

- a. Students develop/use mathematical models (for example: trends, averages, graphs of the number of organisms per unit area in a stable system, etc.) to identify and describe the components that are relevant to support, revise, or refute an explanation about factors affecting biodiversity and ecosystems, including:
 - i. data on numbers and types of organisms are represented and/or
 - ii. interactions between ecosystems at different scales (for example: micro, meso, biome) are represented.
- b. Students support, revise, or refute an explanation of factors affecting biodiversity and population levels, including:
 - i. The populations and number of organisms in ecosystems vary as a function of the physical and biological dynamics of the ecosystem.
 - ii. The response of an ecosystem to a small change might not significantly affect populations, whereas the response to a large change can have a large effect on populations and then feeds back to the ecosystem at a range of scales.
 - iii. Ecosystems can exist in the same location on a variety of scales (for example: plants and animals vs. microbes), and these populations can interact in ways that significantly change these ecosystems (for example: interactions among microbes, plants, and animals can be an important factor in the resources available to both a microscopic and macroscopic ecosystem).

2. Mathematical modeling

- a. Students develop/use a mathematical model (for example: trends, averages, graphs, etc.) of factors affecting biodiversity and ecosystems to identify changes over time in the numbers and types of organisms in ecosystems of different scales.

3. Analysis

- a. Students develop/use and analyze a mathematical model of factors affecting biodiversity and ecosystems:
 - i. to identify the most important factors that determine biodiversity and population numbers of an ecosystem,
 - ii. as evidence to support, revise, or refute explanations for the effects of both abiotic and biotic factors on biodiversity and population size,
 - iii. as evidence to support, revise, or refute explanations for the interactions of ecosystems on different scales, and/or
 - iv. to describe how factors affecting ecosystems at one scale can cause observable changes in ecosystems at a different scale.
- b. Students develop/use a mathematical model to support, revise, or refute explanations for the effects of modest to extreme disturbances on an ecosystem's capacity to return to its original status or become a different ecosystem.
- c. Students develop/use a mathematical model to support, revise, or refute explanations based on new evidence about any factors that affect biodiversity and populations for example: data illustrating the effect of a disturbance within the ecosystem).

B-LS2-2 Academic Language

Question/Sentence Stems

- The quantity of _____ and _____ can be compared.
- The proportion of _____ is _____ because _____.
- The scale of the model of _____ is _____ compared to the actual population.
- To understand the phenomenon of _____, I/we can use a scale of _____ in my/our model.
- How do the quantity of __ and __ compare?
- The effect of __ is __ because _____.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- abiotic
- abundance
- biodiversity
- biotic
- carrying capacity
- climate change
- commensalism
- competition
- deforestation
- density
- disturbance
- equilibrium
- extinction
- feedback
- fluctuation
- generation
- limiting factor
- limiting resource
- mutualism
- parasitism
- pollution
- population
- population control
- predation
- stability
- sustainable
- symbiosis
- trend

B-LS2-3. Construct and revise an explanation based on evidence for the cycling of matter and flow of energy in aerobic and anaerobic conditions.

Clarification Statement: Emphasis is on conceptual understanding of the role of aerobic and anaerobic respiration in the conservation of matter and flow of energy into, out of, and within various ecosystems.

State Assessment Boundary: Assessment focuses on the conceptual understanding and does not include the specific chemical processes of either aerobic or anaerobic respiration.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Constructing Explanations and Designing Solutions Constructing explanations and designing solutions in 9-12 builds on K-8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific ideas, principles, and theories. Construct and revise an explanation based on valid and reliable evidence obtained from a variety of sources (including [models, peer review, simulations, theories, students' own investigations]) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future.	LS2.B: Cycles of Matter and Energy Transfer in Ecosystems Photosynthesis and cellular respiration (including anaerobic processes) provide most of the energy for life processes.	Energy and Matter Energy drives the cycling of matter within and between systems.

Observable features of student performance by the end of the course:

1. Articulating the explanation of phenomena

- a. Students articulate a statement that describes/explains that:
 - i. life depends on biogeochemical cycles (for example: carbon, nitrogen, water) and
 - ii. biogeochemical cycling is driven by aerobic and anaerobic cellular processes.

2. Evidence

- a. Students identify and describe evidence necessary to construct an explanation, including:
 - i. all organisms take in matter and rearrange the atoms in chemical reactions:
 1. photosynthesis captures light energy to create chemical products (limited to glucose) that can be used in cellular respiration and
 2. cellular respiration is the process by which glucose reacts chemically with other compounds (limited to glycolysis, Kreb's cycle, electron transport chain, fermentation), rearranging the matter to release stored chemical energy that is used for essential life processes,
 - ii. biological availability and reservoirs, and
 - iii. key processes of:
 1. Carbon cycle:
 - a. photosynthesis,
 - b. cellular respiration, and
 - c. burning fossil fuels,
 2. Water cycle:
 - a. condensation,
 - b. evaporation,
 - c. precipitation, and
 - d. transpiration,
 3. Nitrogen cycle:
 - a. nitrogen fixation,
 - b. lightning,
 - c. volcanic activity, and
 - d. agricultural production.

3. Reasoning

- a. Students use the following chain of reasoning to connect the evidence and support, refute, or revise an explanation:
 - i. Energy inputs into cells occur either by photosynthesis or taking in matter.
 - ii. All cells engage in cellular respiration (aerobic or anaerobic) and produce products of respiration.
 - iii. The cycling of matter into and out of cells/organisms is driven by the energy captured by photosynthesis and released by respiration.
 - iv. The cycling of matter and transfer of energy must occur whether respiration is aerobic or anaerobic.
 - v. The transfer of energy through respiration also drives the cycling of nutrients between organisms and the environment (for example: nitrogen and water cycles).
- b. Given new data or information, students review the explanation and justify the revision (for example: recent discoveries of life surrounding deep sea ocean vents have shown that photosynthesis is not the only driver for the cycling of matter and transfer of energy in ecosystems).

B-LS2-3 Academic Language

Question/Sentence Stems

- _____ happens to matter as it moves within the system.
- In this system, energy is entering by _____, doing _____ in the system, and leaving the system by _____.
- The energy for _____ is from _____.
- The flow of energy causes _____ to occur in the system.
- The energy is entering the system by _____.
- In the system, the cycling of matter _____.
- The matter in the system enters from _____.
- When the matter leaves the system, it goes _____.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- aerobic
- agricultural production
- anaerobic
- autotroph
- biomass
- carbon
- condensation
- consumer
- cycling of matter
- decomposer
- energy flow
- energy pyramid
- evaporation
- fermentation
- food chain
- food web
- fossil fuel
- glucose
- heterotroph
- lightning
- microbe
- nitrogen
- nitrogen fixation
- photosynthesis
- precipitation
- predator-prey relationship
- producer
- pyramid of biomass
- reservoir
- respiration
- sink
- solar energy
- transpiration
- trophic level
- volcanic activity
- water

B-LS2-4. Use mathematical representations to support claims for the cycling of matter and flow of energy among organisms in an ecosystem.

Clarification Statement: Emphasis is on using a mathematical model of stored energy in biomass to describe the transfer of energy from one trophic level to another and that matter and energy are conserved as matter cycles and energy flows through ecosystems. Emphasis is on conservation of carbon, oxygen, hydrogen, and nitrogen as they move through an ecosystem.

State Assessment Boundary: Assessment is limited to proportional reasoning to describe the cycling of matter and flow of energy.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Using Mathematics and Computational Thinking Mathematical and computational thinking in 9-12 builds on K-8 experiences and progresses to using algebraic thinking and analysis, a range of linear and nonlinear functions including trigonometric functions, exponentials and logarithms, and computational tools for statistical analysis to analyze, represent, and model data. Simple computational simulations are created and used based on mathematical models of basic assumptions. Use mathematical representations of phenomena or design solutions to support claims.	LS2.B: Cycles of Matter and Energy Transfer in Ecosystems Plants or algae form the lowest level of the food web. At each link upward in a food web, only a small fraction of the matter consumed at the lower level is transferred upward, to produce growth and release energy in cellular respiration at the higher level. Given this inefficiency, there are generally fewer organisms at higher levels of a food web. Some matter reacts to release energy for life functions, some matter is stored in newly made structures, and much is discarded. The chemical elements that make up the molecules of organisms pass through food webs and into and out of the atmosphere and soil, and they are combined and recombined in different ways. At each link in an ecosystem, matter and energy are conserved.	Energy and Matter Energy cannot be created or destroyed—it only moves between one place and another place, between objects and/or fields, or between systems.

Observable features of student performance by the end of the course:

1. Representation

- a. Students develop/use a mathematical model to identify and describe the components that are relevant to support, revise, or refute a claim, including relative quantities related to:
 - i. organisms,
 - ii. matter,
 - iii. energy, and
 - iv. food webs in an ecosystem.

2. Mathematical modeling

- a. Students develop/use a mathematical model to describe how a claim can be expressed as a mathematical relationship.
- b. Students develop/use mathematical models of the food web to:
 - i. explain the transfer of matter (as atoms and molecules) and energy upward between organisms and their environment,
 - ii. identify the transfer of energy and matter between trophic levels, and/or
 - iii. identify the relative proportion of organisms at each trophic level by correctly identifying:
 - 1. producers as the lowest trophic level and having the greatest biomass and energy and
 - 2. consumers decreasing in numbers at higher trophic levels.

3. Analysis

- a. Students develop/use mathematical models to support, revise, or refute claims including:
 - i. matter cycles between organisms and their environment and
 - ii. energy transfers from one trophic level to another and through the environment.
- b. Students develop/use and analyze a mathematical model to account for:
 - i. energy not transferred to higher trophic levels, but which is instead used for growth, maintenance, or repair and/or transferred to the environment and/or
 - ii. inefficiencies in transfer of matter and energy.

B-LS2-4 Academic Language

Question/Sentence Stems

- _____ happens to matter as it moves within the system.
- In this system, energy is entering by _____, doing _____ in the system, and leaving the system by _____.
- The energy for _____ is from _____.
- The flow of energy causes _____ to occur in the system.
- The energy is entering the system by _____.
- In the system, the cycling of matter _____.
- The matter in the system enters from _____.
- When the matter leaves the system, it goes _____.
- According to the data, _____ leads to a change in _____.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | |
|---------------------|------------------------------|
| • aerobic | • glucose |
| • anaerobic | • heterotroph |
| • autotroph | • microbe |
| • biomass | • photosynthesis |
| • carbon | • predator-prey relationship |
| • consumer | • producer |
| • cycling of matter | • pyramid of biomass |
| • decomposer | • reservoir/sink |
| • energy flow | • respiration |
| • energy pyramid | • solar energy |
| • food chain | • sugar |
| • food web | • trophic level |
| • fossil fuel | • water |

B-LS2-6. Evaluate claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms *in stable conditions but changing conditions* may result in a new ecosystem.

Clarification Statement: Examples of changes in ecosystem conditions could include modest biological or physical changes, such as moderate hunting or a seasonal flood; and extreme changes, such as volcanic eruption or sea level rise.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Engaging in Argument from Evidence Engaging in argument from evidence in 9-12 builds on K-8 experiences and progresses to using appropriate and sufficient evidence and scientific reasoning to defend and critique claims and explanations about the natural and designed world(s). Arguments may also come from current scientific or historical episodes in science. Evaluate the claims, evidence, and reasoning behind currently accepted explanations or solutions to determine the merits of arguments.	LS2.C: Ecosystem Dynamics, Functioning, and Resilience A complex set of interactions within an ecosystem can keep its numbers and types of organisms relatively constant over long periods of time under stable conditions. If a modest biological or physical disturbance to an ecosystem occurs, it may return to its more or less original status (i.e., the ecosystem is resilient), as opposed to becoming a very different ecosystem. Extreme fluctuations in conditions or the size of any population, however, can challenge the functioning of ecosystems in terms of resources and habitat availability.	Stability and Change Much of science deals with constructing explanations of how things change and how they remain stable.

*Because populations can be large, students may need to understand and apply the conventions of scientific notation when working with quantities of measurement and their calculations.

Observable features of student performance by the end of the course:

1. Identifying the supporting claims, evidence, and reasoning

- a. Students identify the explanation that complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem that is supported by the claims, evidence, and reasoning, including:
 - i. the claims to be evaluated,
 - ii. the evidence to be evaluated, and
 - iii. the reasoning to be evaluated.

2. Identifying any additional evidence relevant to the evaluation

- a. Students identify and describe additional evidence (for example: data, information, etc.) that was not provided but is relevant to the explanation and its evaluation, including:
 - i. factors that affect biodiversity,
 - ii. relationships between species and the physical environment in an ecosystem, and/or
 - iii. changes in the numbers of species and organisms in an ecosystem that has been subjected to a modest or extreme change in ecosystem conditions.

3. Evaluating and critiquing

- a. Students identify and describe the strengths and weaknesses of the explanation by accurately describing a particular response of biodiversity to a changing condition, based on an understanding of the factors that affect biodiversity and the relationships between species and the physical environment in an ecosystem.
- b. Students use the additional evidence to assess the validity and reliability of the explanation's evidence and its ability to support the argument that resiliency of an ecosystem is subject to the degree of change in the biological and physical environment.
- c. Students assess the logic of the reasoning, including the relationship between degree of change and stability in ecosystems, and the utility of the reasoning in supporting the explanation of how:
 - i. Modest biological or physical disturbances in an ecosystem result in maintenance of relatively consistent numbers and types of organisms.
 - ii. Extreme fluctuations in conditions of the size of any population can challenge the functioning of ecosystems in terms of resources and habitat availability and can even result in a new ecosystem.

B-LS2-6 Academic Language

Question/Sentence Stems

- The key components of the system are_____.
- In the system, _____ and _____ are shown in the model.
- In the system, _____ and _____ work together to_____.
- In the system, _____ and _____ interact in _____ way.
- In this system, if ___ disturbance occurs, _____ results.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | |
|--------------------|------------------------|
| • anthropogenic | • human impact |
| • biodiversity | • intermediate species |
| • biosphere | • niche |
| • carbon cycle | • nitrogen cycle |
| • climate change | • organism |
| • climax community | • pioneer species |
| • consumer | • primary succession |
| • decomposer | • producer |
| • disturbance | • resources |
| • ecosystem | • secondary succession |
| • environment | • species |
| • equilibrium | • stability |
| • extinction | • succession |
| • fluctuation | • water cycle |
| • habitat | |

B-LS2-8. Evaluate evidence for the role of group behavior on individual and species' chances to survive and reproduce.

Clarification Statement: Emphasis is on: (1) distinguishing between group and individual behavior, (2) identifying evidence supporting the outcomes of group behavior, and (3) developing logical and reasonable arguments based on evidence. Examples of group behaviors could include flocking, schooling, or herding, and cooperative behaviors such as hunting, migrating, or swarming.

Science and Engineering Practices	Disciplinary Core Ideas	<i>Crosscutting Concepts</i>
Engaging in Argument from Evidence Engaging in argument from evidence in 9-12 builds on K-8 experiences and progresses to using appropriate and sufficient evidence and scientific reasoning to defend and critique claims and explanations about the natural and designed world(s). Arguments may also come from current scientific or historical episodes in science. Evaluate the evidence behind currently accepted explanations to determine the merits of arguments.	LS2.D: Social Interactions and Group Behavior Group behavior has evolved because membership can increase the chances of survival for individuals and their genetic relatives.	Cause and Effect Empirical evidence is required to differentiate between cause and correlation and make claims about specific causes and effects.

Observable features of student performance by the end of the course:

- 1. Identifying the given explanation and the supporting evidence**
 - a. Students identify an explanation that group behavior can increase the chances for an individual and species to survive and reproduce that is supported by the evidence to be evaluated.
- 2. Identifying additional evidence relevant to the evaluation**
 - a. Students identify and describe additional evidence (for example: data, information, etc.) that was not provided but is relevant to the explanation and its evaluation including:
 - i. relationships between specific group behaviors (for example: flocking, schooling, herding, cooperative hunting, migrating, swarming) and individual fitness.

3. Evaluating and critiquing

- a. Students use additional evidence to assess the validity, reliability, strengths, and weaknesses of the given evidence and evaluate its ability to support logical and reasonable arguments about the outcomes of group behavior.
- b. Students evaluate the given evidence for the degree to which it supports an explanation that group behavior can have a survival advantage for some species including:
 - i. how it supports cause and effect relationships between various kinds of group behavior and individual survival rates.

B-LS2-8 Academic Language

Question/Sentence Stems

- By looking at patterns in the data, I/we determined that _____ caused _____.
- _____ caused the patterns I am observing. I know this because _____.
- If _____ happens, I/we predict that _____ will occur.
- Even though I/we cannot see _____, it explains why _____ is happening.
- When I/we change _____ in the system, _____ is affected.
- The probability that _____ caused _____ is _____. I/We know this because _____.
- The evidence _____ presented in the scenario supports the claim that _____ causes _____.
- To conclude that _____ caused _____, the following evidence is needed _____.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | |
|-----------------------|-----------------------|
| • altruism | • natural selection |
| • behavior | • predator |
| • beneficial | • prey |
| • competition | • proliferation |
| • cooperative hunting | • schooling |
| • evolution | • selective pressure |
| • fitness | • sexual reproduction |
| • flocking | • social interaction |
| • herding | • species |
| • limited resource | • swarming |
| • migration | |

LS3 – Heredity: Inheritance and Variation of Traits

B-LS3-1. Ask questions to clarify relationships about the role of DNA and chromosomes in coding the instructions for characteristic traits passed from parents to offspring.

State Assessment Boundary: Assessment does not include the phases of meiosis or the biochemical mechanism of specific steps in the process (including gene regulation).

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Asking Questions and Defining Problems Asking questions and defining problems in 9- 12 builds on K-8 experiences and progresses to formulating, refining, and evaluating empirically testable questions and design problems using models and simulations. Ask questions that arise from examining models or a theory to clarify relationships.	LS1.A: Structure and Function All cells contain genetic information in the form of DNA molecules. Genes are regions in the DNA that contain the instructions that code for the formation of proteins. (<i>secondary</i>) LS3.A: Inheritance of Traits Each chromosome consists of a single very long DNA molecule, and each gene on the chromosome is a particular segment of that DNA. The instructions for forming species' characteristics are carried in DNA. All cells in an organism have the same genetic content, but the genes used (expressed) by the cell may be regulated in different ways. Not all DNA codes for a protein; some segments of DNA are involved in regulatory or structural functions, and some have no as-yet known function.	Cause and Effect Empirical evidence is required to differentiate between cause and correlation and make claims about specific causes and effects.

*Students will need to examine and interpret a variety of models (for example: pedigree, Punnett Square, karyotype, etc.).

*Because genome sizes, mutation rates, gene frequencies, etc. can be large, students may need to understand and apply the conventions of scientific notation when working with quantities of measurement and their calculations.

Observable features of student performance by the end of the course:

1. Addressing phenomena of the natural world

- a. Students develop/use models of DNA to formulate questions to clarify:
 - i. the cause-and-effect relationships between DNA, the proteins it codes for, phenotypic expression,
 - ii. that DNA and chromosomes can be regulated in multiple ways (for example: differentiation), and/or
 - iii. the relationship between the non-protein coding sections of DNA and their functions (for example: regulatory functions) in organisms.

2. Evaluating empirical testability

- a. Student questions are empirically testable.

B-LS3-1 Academic Language

Question/Sentence Stems

- By looking at patterns in the data, I/we determined that _____ caused _____.
- _____ caused the patterns I am observing. I know this because _____.
- If _____ happens, I/we predict that _____ will occur.
- Even though I/we cannot see _____, it explains why _____ is happening.
- When I/we change _____ in the system, _____ is affected.
- The probability that _____ caused _____ is _____. I/We know this because _____.
- The evidence _____ presented in the scenario supports the claim that _____ causes _____.
- To conclude that _____ caused _____, the following evidence is needed _____.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | | |
|-------------------|----------------|------------------|
| • allele | • heterozygous | • pattern |
| • chromosomes | • homozygous | • pedigree |
| • codominant | • incomplete | • phenotype |
| • dominant | dominance | • probability |
| • epigenetics | • inheritance | • Punnett square |
| • exon | • intron | • recessive |
| • gene expression | • Karyotype | • sex-linked |
| • genes | • methylation | • trait |
| • genotype | • offspring | |
| • heredity | • parent | |

B-LS3-2. Make and defend a claim based on evidence that inheritable genetic variations *may result from* (1) new genetic combinations through meiosis, (2) viable errors occurring during replication, and/or (3) mutations caused by environmental factors.

Clarification Statement: Emphasis is on using data to support arguments for the way genetic variation occurs.

State Assessment Boundary: Assessment does not include the phases of meiosis or the biochemical mechanism of specific steps in the process.

Science and Engineering Practices	Disciplinary Core Ideas	<i>Crosscutting Concepts</i>
Engaging in Argument from Evidence Engaging in argument from evidence in 9-12 builds on K-8 experiences and progresses to using appropriate and sufficient evidence and scientific reasoning to defend and critique claims and explanations about the natural and designed world(s). Arguments may also come from current scientific or historical episodes in science. Make and defend a claim based on evidence about the natural world that reflects scientific knowledge, and student-generated evidence.	LS3.B: Variation of Traits In sexual reproduction, chromosomes can sometimes swap sections during the process of meiosis (cell division), thereby creating new genetic combinations and thus more genetic variation. Although DNA replication is tightly regulated and remarkably accurate, errors do occur and result in mutations, which are also a source of genetic variation. Environmental factors can also cause mutations in genes, and viable mutations are inherited. Environmental factors also affect expression of traits, and hence affect the probability of occurrences of traits in a population. Thus, the variation and distribution of traits observed depends on both genetic and environmental factors.	Cause and Effect Empirical evidence is required to differentiate between cause and correlation and make claims about specific causes and effects.

*References to “viable errors occurring during replication” are operationally defined as errors that bypass DNA proofreading (the cell cycle successfully moves past G₂).

Observable features of student performance by the end of the course:

1. Developing a claim

- a. Students make a claim that inheritable genetic variations may result from:
 - i. new genetic combinations through meiosis,
 - ii. viable errors occurring during replication, and/or
 - iii. mutation caused by environmental factors (mutagen).

2. Identifying scientific evidence

- a. Students identify and describe the evidence that supports the claim, including:
 - i. Variation in genetic material naturally results during meiosis from
 - 1. crossing over and
 - 2. independent assortment.
 - ii. Genetic mutations can occur due to
 - 1. replication errors and
 - 2. environmental factors (mutagen).
 - iii. Genetic material is inheritable.

3. Evaluating and critiquing evidence

- a. Students identify strengths and weaknesses of the evidence used to support the claim, including:
 - i. types of sources,
 - ii. sufficiency, including validity and reliability, of the evidence to make and defend the claim, and/or
 - iii. any alternative interpretations of the evidence and why the evidence supports or does not support the student's claim, as opposed to any other claims.

4. Reasoning and synthesis

- a. Students use the following chain of reasoning to connect the evidence:
 - i. Genetic mutations produce genetic variation between cells or organisms.
 - ii. Genetic variations produced by mutation and meiosis can be inherited.
- b. Students use reasoning and evidence to explain that new combinations of DNA can arise from several sources, including:
 - i. meiosis,
 - ii. replication error, and/or
 - iii. mutations caused by mutagens.
- c. Students defend a claim against counterclaims by evaluating counterclaims and by describing the connections between the relevant and appropriate evidence to the strongest claim.

B-LS3-2 Academic Language

Question/Sentence Stems

- By looking at patterns in the data, I/we determined that _____ caused _____.
- _____ caused the patterns I am observing. I know this because _____.
- If _____ happens, I/we predict that _____ will occur.
- Even though I/we cannot see _____, it explains why _____ is happening.
- When I/we change _____ in the system, _____ is affected.
- The probability that _____ caused _____ is _____. I/We know this because _____.
- The evidence _____ presented in the scenario supports the claim that _____ causes _____.
- To conclude that _____ caused _____, the following evidence is needed _____.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | |
|-------------------------------|--------------------------|
| • allele | • homologous chromosome |
| • cellular division | • independent assortment |
| • centromere | • inherited |
| • chromatid | • insertion |
| • chromosome | • meiosis |
| • codon (chart) | • meiosis I |
| • crossing over | • meiosis II |
| • daughter cell | • monosomy |
| • deletion | • multicellular |
| • diploid | • mutagen |
| • deoxyribonucleic acid (DNA) | • mutation |
| • egg cell | • nondisjunction |
| • epigenetic | • offspring |
| • fertilization | • parent cell |
| • frameshift | • point mutation |
| • gamete | • replication |
| • gene | • sexual reproduction |
| • gene expression | • somatic cell |
| • gene mutation | • sperm cell |
| • genetic code | • substitution |
| • genetic variation | • trait |
| • genome | • trisomy |
| • haploid | |

B-LS3-3. Apply concepts of statistics and probability to explain the variation and distribution of expressed traits in a population.

Clarification Statement: Emphasis is on the use of mathematics to describe the probability of traits as it relates to genetic and environmental factors in the expression of traits.

State Assessment Boundary: Assessment does not include Hardy-Weinberg or Chi-square analysis.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Analyzing and Interpreting Data Analyzing data in 9-12 builds on K-8 experiences and progresses to introducing more detailed statistical analysis, the comparison of data sets for consistency, and the use of models to generate and analyze data. Apply concepts of statistics and probability (including determining function fits to data, slope, intercept, and correlation coefficient for linear fits) to scientific and engineering questions and problems, using digital tools when feasible.	LS3.B: Variation of Traits Environmental factors also affect expression of traits, and hence affect the probability of occurrences of traits in a population. Thus, the variation and distribution of traits observed depends on both genetic and environmental factors.	Scale, Proportion, and Quantity Algebraic thinking is used to examine scientific data and predict the effect of a change in one variable on another (e.g., linear growth vs. exponential growth).

Observable features of student performance by the end of the course:

1. Organizing data

- a. Students organize data for expressed traits in the population by:
 - i. distribution,
 - ii. frequency, and/or
 - iii. variation.

2. Identifying relationships

- a. Students analyze data (for example: probability measures, Hardy-Weinberg Conditions) to determine the relationship between a trait's occurrence within a population and environmental factors.

3. Interpreting data

- a. Students analyze and interpret data to explain the distribution of expressed traits, including:
 - i. Recognition and use of patterns in statistical analysis to predict changes in trait distribution within a population if environmental variables change.
 - ii. Description of the expression of a chosen trait and its relationship to some environmental factors based on reliable evidence.

B-LS3-3 Academic Language

Question/Sentence Stems

- How can the quantity of _____ and _____ can be compared?
- The proportion of _____ is _____ because _____.
- The scale of the model of _____ is _____ compared to the actual population.
- To understand the phenomenon of _____, I/we can use a scale of _____ in my/our model.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | |
|----------------------------------|-------------------------|
| • allele | • heterozygous |
| • autosomal | • homozygous |
| • codominance | • incomplete dominance |
| • complete dominance | • karyotype |
| • dihybrid cross | • monohybrid cross |
| • distribution | • P (parental) |
| • dominant | • pedigree |
| • environmental | • phenotype |
| • F ₁ (first filial) | • phenotypic ratio |
| • F ₂ (second filial) | • polygenic inheritance |
| • gamete | • population |
| • gene expression | • probability |
| • genetic variability | • Punnett square |
| • genotype | • ratio |
| • genotypic ratio | • recessive |
| • Hardy-Weinberg Conditions | • sex linked |
| • Hardy-Weinberg Equilibrium | • trait |
| • heredity | • variation |

LS4 – Biological Evolution: Unity and Diversity

B-LS4-1. Communicate scientific information that common ancestry and biological evolution *are supported by multiple lines of empirical evidence.*

Clarification Statement: Emphasis is on students’ conceptual understanding of the role each line of evidence has relating to common ancestry and biological evolution. Examples of evidence could include similarities in DNA sequences, anatomical structures, and order of appearance of structures in embryological development.

State Assessment Boundary: Assessment is limited to conceptual explanations of the evidence for biological evolution and is not extended to the lines of evidence for specific species. Assessment does not include classification of organisms.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Obtaining, Evaluating, and Communicating Information Obtaining, evaluating, and communicating information in 9-12 builds on K-8 experiences and progresses to evaluating the validity and reliability of the claims, methods, and designs. Communicate scientific information (e.g., about phenomena and/or the process of development and the design and performance of a proposed process or system) in multiple formats (including orally, graphically, textually, and mathematically).	LS4.A: Evidence of Common Ancestry and Diversity Genetic information, like the fossil record, provides evidence of evolution. DNA sequences vary among species, but there are many overlaps; notably, the ongoing branching that produces multiple lines of descent can be inferred by comparing the DNA sequences of different organisms. Such information is also derivable from the similarities and differences in amino acid sequences and from anatomical and embryological evidence. ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World The understanding of evolutionary relationships has recently been greatly accelerated by using new molecular tools to study biology.	Patterns Different patterns may be observed at each of the scales at which a system is studied and can provide evidence for causality in explanations of phenomena.

*Because the different lines of evidence have varying scales, students may need to understand and apply the conventions of scientific notation when working with quantities of measurement and their calculations.

Observable features of student performance by the end of the course:

1. Communication

- a. Students use and cite at least two different formats (for example: oral, graphical, textual, mathematical, etc.) to communicate scientific information about how common ancestry and biological evolution are supported by multiple lines of empirical evidence.

2. Connections

- a. Students identify and communicate evidence for common ancestry and biological evolution, including:
 - i. Information derived from DNA sequences, which vary among species but have many similarities between species.
 - ii. Similarities of the patterns or amino acid sequences, even when DNA sequences are slightly different, including that multiple patterns of DNA sequences can code for the same amino acids.
 - iii. The triplet codon pattern (for example: codon chart) that is used to decode sequences of mRNA for protein construction is universal amongst known species.
 - iv. Patterns in the fossil record (for example: presence, location, and possible inferences in lines of evolutionary descent for multiple specimen [biogeography]).
 - v. Patterns of anatomical and embryological similarities.
- b. Students identify and communicate connections between the lines of evidence and the claim of common ancestry and biological evolution.
- c. Students communicate that together, the patterns observed at multiple spatial and temporal scales (for example: DNA sequences, embryological development, fossil record) provide evidence for relationships relating biological evolution and common ancestry.

B-LS4-1 Academic Language

Question/Sentence Stems

- I/We can observe (notice) the pattern of _____ presented in the data collected.
- I/We can observe (notice) the pattern of _____ in the data presented.
- The pattern seen in the collected data allows me/us to conclude (know) that _____.
- The observed pattern supports the conclusion that _____ is caused by _____, because _____.
- The pattern of _____ is changing over time.
- The following predictions can be made about _____ when using the pattern of _____ found in the data.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- adaptation
- amino acid sequencing
- analogous structure
- anatomy
- biochemical evidence
- biogeography
- cladogram
- common ancestry
- comparative anatomy
- descent with modification
- DNA sequencing
- electrophoresis
- embryo
- embryology
- evolution
- evolutionary tree
- fossil record
- heritable trait
- homologous structure
- homology
- natural selection
- paleontology
- phenotypic similarity
- phylogenetic tree
- phylogeny
- sedimentary layers
- species
- vestigial structure

B-LS4-3. Apply concepts of statistics and probability to support explanations that organisms with an advantageous heritable trait *tend to* increase in proportion to organisms lacking this trait.

Clarification Statement: Emphasis is on analyzing shifts in numerical distribution of traits and using these shifts as evidence to support explanations.

State Assessment Boundary: Assessment is limited to basic statistical and graphical analysis. Assessment does not include allele frequency calculations.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Analyzing and Interpreting Data Analyzing data in 9-12 builds on K-8 experiences and progresses to introducing more detailed statistical analysis, the comparison of data sets for consistency, and the use of models to generate and analyze data. Apply concepts of statistics and probability (including determining function fits to data, slope, intercept, and correlation coefficient for linear fits) to scientific and engineering questions and problems, using digital tools when feasible.	LS4.B: Natural Selection Natural selection occurs only if there is both (1) variation in the genetic information between organisms in a population and (2) variation in the expression of that genetic information—that is, trait variation—that leads to differences in performance among individuals. The traits that positively affect survival are more likely to be reproduced, and thus are more common in the population. LS4.C: Adaptation Natural selection leads to adaptation, that is, to a population dominated by organisms that are anatomically, behaviorally, and physiologically well suited to survive and reproduce in a specific environment. That is, the differential survival and reproduction of organisms in a population that have an advantageous heritable trait leads to an increase in the proportion of individuals in future generations that have the trait and to a decrease in the proportion of individuals that do not. Adaptation also means that the distribution of traits in a population can change when conditions change.	Patterns Different patterns may be observed at each of the scales at which a system is studied and can provide evidence for causality in explanations of phenomena.

*Because populations can be large, students may need to understand and apply the conventions of scientific notation when working with quantities of measurement and their calculations.

Observable features of student performance by the end of the course:

1. Organizing data

- a. Students organize data (for example: tables, charts, graphs, etc.) by the distribution of genetic traits over time.

2. Identifying relationships

- a. Students analyze data (for example: probability measures) to determine patterns of change in numerical distributions of traits over various time and population scales.

3. Interpreting data

- a. Students use the analyzed data as evidence to support explanations about:
 - i. Positive or negative effects on fitness of individuals as relating to their expression of a variable trait in a population.
 - ii. Natural selection as the cause of increases and decreases in heritable traits over time in a population, but only if it affects reproductive success.
 - iii. The changes in distribution of adaptations of anatomical, behavioral, and physiological traits in a population.

B-LS4-3 Academic Language

Question/Sentence Stems

- I/We can observe (notice) the pattern of _____ presented in the data collected.
- I/We can observe (notice) the pattern of _____ in the data presented.
- The pattern seen in the collected data allows me/us to conclude (know) that _____.
- The observed pattern supports the conclusion that _____ is caused by _____, because _____.
- The pattern of _____ is changing over time.
- The observed trait _____ is changing in the population by a noticed _____ in the model.
- The following predictions can be made about _____ when using the pattern of _____ found in the data.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- adaptation
- advantageous
- allele
- distribution
- fitness
- gene
- heritable trait
- increase
- natural selection
- proportion
- significance
- variation

B-LS4-6. Create or revise a simulation to test a solution to mitigate adverse *impacts of* human activity on biodiversity.

Clarification Statement: Emphasis is on testing solutions for a proposed problem related to threatened or endangered species, or to genetic variation of organisms for multiple species.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Using Mathematics and Computational Thinking Mathematical and computational thinking in 9-12 builds on K-8 experiences and progresses to using algebraic thinking and analysis, a range of linear and nonlinear functions including trigonometric functions, exponentials and logarithms, and computational tools for statistical analysis to analyze, represent, and model data. Create or revise a simulation of a phenomenon, designed device, process, or system.	LS4.C: Adaptation Changes in the physical environment, whether naturally occurring or human induced, have thus contributed to the expansion of some species, the emergence of new distinct species as populations diverge under different conditions, and the decline — and sometimes the extinction — of some species. LS4.D: Biodiversity and Humans Humans depend on the living world for the resources and other benefits provided by biodiversity. But human activity is also having adverse impacts on biodiversity through overpopulation, overexploitation, habitat destruction, pollution, introduction of invasive species, and climate change. Thus, sustaining biodiversity so that ecosystem functioning and productivity are maintained is essential to supporting and enhancing life on Earth. Sustaining biodiversity also aids humanity by preserving landscapes of recreational or inspirational value. ETS1.B: Developing Possible Solutions When evaluating solutions, it is important to take into account a range of constraints, including cost, safety, reliability, and aesthetics, and to consider social, cultural, and environmental impacts. (<i>secondary</i>) Both physical models and computers can be used in various ways to aid in the engineering design process. Computers are useful for a variety of purposes, such as running simulations to test different ways of solving a problem or to see which one is most efficient or economical; and in making a persuasive presentation to a client about how a given design will meet his or her needs. (<i>secondary</i>) ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World Modern civilization depends on major technological systems, including those related to agriculture, health, water, energy, transportation (e.g., wildlife corridors), manufacturing, construction, and communications. (<i>secondary</i>) Engineers continuously modify these technological systems by applying scientific knowledge and engineering design practices to increase benefits while decreasing costs and risks. (<i>secondary</i>)	Cause and Effect Empirical evidence is required to differentiate between cause and correlation and make claims about specific causes and effects.

Observable features of student performance by the end of the course:

1. Representation

- a. Students create or revise a simulation that:
 - i. Models effects of human activity on threatened or endangered species or to the genetic variation within a species. Examples of human activity include:
 - 1. adverse habitat alterations,
 - 2. climate change,
 - 3. invasive species,
 - 4. overexploitation,
 - 5. overpopulation, and/or
 - 6. pollution.
 - ii. Provides quantitative information about the effect of a solution on threatened or endangered species.
- b. Students describe the components that are modeled by the simulation, including human activity and the factors that affect biodiversity.
- c. Students describe the variables that can be changed by the user to evaluate the proposed solutions, tradeoffs, or other decisions.

2. Computational Modeling

- a. Students use logical and realistic inputs for the simulation that show an understanding of the reliance of ecosystem function and productivity on biodiversity, and that consider the following restraints:
 - i. Cost,
 - ii. cultural and environmental impacts,
 - iii. reliability, and/or
 - iv. safety.
- b. Students use the simulation to identify possible negative consequences of solutions that would outweigh their benefits.

3. Analysis

- a. Students compare the simulation results to expected results.
- b. Students analyze the simulation results to determine whether the simulation provides sufficient information to evaluate the solution.
- c. Students identify the limits of the simulation.
- d. Students interpret simulation results and predict the effects of the specific design solutions on biodiversity based on that interpretation.

4. Revision

- a. Students revise the simulation to provide sufficient information to evaluate the simulation.

B-LS4-6 Academic Language

Question/Sentence Stems

- By looking at patterns in the data, I/we determined that _____ caused _____.
- _____ caused the patterns I am observing. I know this because _____.
- If _____ happens, I/we predict that _____ will occur.
- Even though I/we cannot see _____, it explains why _____ is happening.
- The probability that _____ caused _____ is _____. I/We know this because _____.
- The evidence _____ presented in the scenario supports the claim that _____ causes _____.
- To conclude that _____ caused _____, the following evidence is needed _____.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | |
|--------------------------|---------------------------|
| • abiotic | • extinction |
| • anthropogenic | • genetic diversity |
| • biodiversity | • habitat |
| • biotic | • habitat destruction |
| • captive breeding | • habitat fragmentation |
| • carrying capacity | • habitat restoration |
| • climate | • human impact |
| • climate change | • human population growth |
| • conservation | • invasive species |
| • constraints | • iterative |
| • criteria | • mitigation |
| • deforestation | • optimize |
| • ecological restoration | • overharvesting |
| • ecosystem | • pollution |
| • ecosystem diversity | • solution |
| • ecotourism | • species diversity |
| • endangered species | • sustainable development |
| • environment | |

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Appendix A: Biology 2 Grade Vertical Articulation of Disciplinary Core Idea Instructional Focus Statements

This appendix presents the vertical articulation of Disciplinary Core Idea (DCI) Aligned Instructional Focus statements found in the Vertical Articulations on the [Science Instructional Resources webpage](#). It is intended to help educators understand how conceptual understanding develops across grade levels and courses and how current learning connects to both prior and future instruction.

This appendix is designed to support instructional planning, curriculum alignment, and understanding of the broader progression of science concepts across grades and courses.

Instructional Focus statements clarify content expectations associated with a Performance Expectation but should be considered alongside the associated Science and Engineering Practices (SEPs) and Crosscutting Concepts (CCCs).

The appendix is organized by Performance Expectation (PE) and associated DCI, using the following format:

Domain Core Idea

Performance Expectation

Disciplinary Core Idea

Previous Learning	Current Course or Content
Related DCI Aligned Instructional Focus statements from earlier grades are presented in this column to illustrate the conceptual foundations of the current learning.	DCI Aligned Instructional Focus statements for the current grade are presented in this column. If a statement aligns to more than one PE, the applicable PE code(s) follow the statement. Only relevant Extended Knowledge statements are included.

The appendix is not intended to represent a complete curriculum or pacing guide. Rather, it provides a conceptual map of how the knowledge and understandings associated with a DCI progress across grades, courses, and content.

LS1 – From Molecules to Organisms: Structures and Processes

B-LS1-1. Construct an explanation based on evidence for how the structure of DNA determines the structure of proteins, which carry out the essential functions of life through systems of specialized cells.

LS1.A: Structure and Function

Biology 1	Biology 2
- Describe the structure and functions of DNA. - Describe proteins as the primary functional products of genetic information that enable cells and tissues to perform specialized functions. - Apply the pathway from DNA in the nucleus to protein production by summarizing transcription to mRNA, polypeptide assembly at ribosomes, and protein modification and packaging by the ER and Golgi apparatus. - Explain that all cells in an organism contain the same DNA but can produce different sets of proteins that give rise to specialized cell functions. - Analyze how changes in a gene's information can change a protein and alter cell or tissue function at a conceptual level.	- Differentiate coding and non-coding DNA and relate non-coding regions to organizational or regulatory roles. (B-LS3-1) - Describe introns and exons in eukaryotic genes and explain how RNA processing produces mature mRNA, allowing a single gene to support multiple functional outcomes. (B-LS3-1) - Outline major stages of transcription and translation and the roles of key cellular structures involved, without reaction mechanisms or kinetics. <i>Extended Knowledge</i> - Evaluate epigenetic influences, for example methylation, on gene activity conceptually.

LS3.A: Inheritance of Traits

Biology 1	Biology 2
- Explain that gene's nucleotide sequence is used to make proteins through transcription of DNA into mRNA and translation of mRNA into a polypeptide. - Explain how traits are inherited and expressed, using central dogma reasoning. - Explain that changes in a gene's sequence can result in changes to a protein and lead to differences in traits.	- Explain that a gene's nucleotide sequence specifies a protein or a functional RNA product. - Explain how changes in DNA sequence can alter proteins or RNAs and thereby influence traits at the organism level. - Identify that some DNA does not code for proteins and plays regulatory or structural roles that influence gene expression.

LS2 – Ecosystems: Interactions, Energy, and Dynamics

B-LS2-2. Use mathematical representations to support and revise explanations based on evidence about factors affecting biodiversity and populations in ecosystems of different scales.

LS2.A: Interdependent Relationships in Ecosystems

Biology 1	Biology 2
- Define carrying capacity as the limit on population size set by finite resources and biotic challenges (predation, competition, disease). - Describe how limits from resources and biotic challenges affect species abundance within defined ecosystem boundaries. - Explain how changes in a limiting factor (for example, decreased prey or increased disease) can shift carrying capacity. - Describe overshoot and subsequent dieback qualitatively, as temporary population increases beyond resource limits followed by reduction toward sustainable levels.	- Describe carrying capacity as a function of finite resources and biotic challenges, and explain how this constrains population abundance. - Explain that abundance reflects the tension between potential population growth and environmental limits. - Differentiate outcomes associated with modest versus large disturbances, relating disturbance magnitude to expected population and ecosystem responses.

LS2.C: Ecosystem Dynamics, Functioning, and Resilience

Biology 1	Biology 2
- Identify human activities that disrupt ecosystems, including habitat destruction, pollution, introduction of invasive species, overexploitation, and long-term changes in climate patterns. - Describe how these activities alter environmental conditions or resource availability within ecosystems. - Explain how such changes can limit organism survival or reproduction, threatening the persistence of some species. - Describe that actions such as habitat restoration, invasive-species management, pollution reduction, and sustainable harvest practices can improve conditions that support ecosystem functioning.	- Describe that under relatively stable conditions, complex interactions can keep ecosystem structure and population sizes within identifiable ranges. (B-LS2-6) - Explain that ecosystems may return to prior states following modest disturbances when key species, resources, and interactions remain intact (resilience). (B-LS2-6) - Explain that extreme disturbances or large population changes can disrupt resources and habitat availability, resulting in ecosystems with different structures and interactions. (B-LS2-6) - Describe factors that influence biodiversity, including resource availability, species interactions, and physical environmental conditions. (B-LS2-6)

B-LS2-3. Construct and revise an explanation based on evidence for the cycling of matter and flow of energy in aerobic and anaerobic conditions.

LS2.B: Cycles of Matter and Energy Transfer in Ecosystems

Biology 1	Biology 2
• Describe the carbon cycle as the movement of carbon among organisms and between the biosphere, atmosphere, hydrosphere, and geosphere. • Explain how photosynthesis, cellular respiration, decomposition, and combustion connect organisms and environments through carbon exchange.	• Explain that photosynthesis provides stored chemical energy that fuels cellular respiration and supports ecosystem-level energy flow. (B-LS2-4) • Describe that all organisms carry out cellular respiration (aerobic or anaerobic) to release chemical energy for life processes and return matter to the environment. (B-LS2-4)

B-LS2-4. Use mathematical representations to support claims for the cycling of matter and flow of energy among organisms in an ecosystem.

LS2.B: Cycles of Matter and Energy Transfer in Ecosystems

Biology 1	Biology 2
- Describe the carbon cycle as the movement of carbon among organisms and between the biosphere, atmosphere, hydrosphere, and geosphere. - Explain how photosynthesis, cellular respiration, decomposition, and combustion connect organisms and environments through carbon exchange.	- Explain that photosynthesis provides stored chemical energy that fuels cellular respiration and supports ecosystem-level energy flow. (B-LS2-3) - Describe that all organisms carry out cellular respiration (aerobic or anaerobic) to release chemical energy for life processes and return matter to the environment. (B-LS2-3) - Explain that as matter and energy move through food webs, only a fraction becomes biomass at each successive trophic level, and this inefficiency limits the number of trophic levels in ecosystems. - Recognize that matter (atoms) is conserved as it moves through food webs and is recombined into different molecules, while energy flows through ecosystems and is released to the environment. <i>Extended Knowledge</i> - Understand that limits on energy transfer help explain common food-web patterns.

B-LS2-6. Evaluate claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions but changing conditions may result in a new ecosystem.

LS2.C: Ecosystem Dynamics, Functioning, and Resilience

Biology 1	Biology 2
- Identify human activities that disrupt ecosystems, including habitat destruction, pollution, introduction of invasive species, overexploitation, and long-term changes in climate patterns. - Describe how these activities alter environmental conditions or resource availability within ecosystems. - Explain how such changes can limit organism survival or reproduction, threatening the persistence of some species. - Describe that actions such as habitat restoration, invasive-species management, pollution reduction, and sustainable harvest practices can improve conditions that support ecosystem functioning.	- Describe that under relatively stable conditions, complex interactions can keep ecosystem structure and population sizes within identifiable ranges. (B-LS2-2) - Explain that ecosystems may return to prior states following modest disturbances when key species, resources, and interactions remain intact (resilience). (B-LS2-2) - Explain that extreme disturbances or large population changes can disrupt resources and habitat availability, resulting in ecosystems with different structures and interactions. (B-LS2-2) - Describe factors that influence biodiversity, including resource availability, species interactions, and physical environmental conditions. (B-LS2-2) - Explain that ecosystem stability depends on both the magnitude of environmental change and the relationships among organisms and conditions. <i>Extended Knowledge</i> - Understand that disruptions at one trophic level can have cascading effects on overall ecosystem functioning.

B-LS2-8. Evaluate evidence for the role of group behavior on individual and species' chances to survive and reproduce.

LS2.D: Social Interactions and Group Behavior

Third Grade	Biology 2
• Describe that some animals form groups that help individuals obtain food, defend themselves, or cope with environmental changes. • Explain that belonging to a group can increase an individual's chance of survival compared with being alone. • Analyze examples of animal groups with different sizes and structures (for example, family groups or groups with dominant members). • Describe that some animal groups remain stable over time, while others change membership.	• Define group behavior as coordinated or cooperative actions among individuals of the same species that can increase survival or reproductive success. • Describe common benefits of group behavior, such as predator avoidance, improved foraging, or energy efficiency. • Explain how group behavior can provide relative fitness advantages to individuals and their genetic relatives compared with solitary behavior. <i>Extended Knowledge</i> • Understand that environmental conditions can influence which group behaviors are advantageous in a given context.

LS3 – Heredity: Inheritance and Variation of Traits

B-LS3-1. Ask questions to clarify relationships about the role of DNA and chromosomes in coding the instructions for characteristic traits passed from parents to offspring.

LS1.A: Structure and Function

Biology 1	Biology 2
- Describe the structure and functions of DNA. - Explain that all cells in an organism contain the same DNA but can produce different sets of proteins that give rise to specialized cell functions. - Analyze how changes in a gene's information can change a protein and alter cell or tissue function at a conceptual level.	- Differentiate coding and non-coding DNA and relate non-coding regions to organizational or regulatory roles. (B-LS1-1) - Describe introns and exons in eukaryotic genes and explain how RNA processing produces mature mRNA, allowing a single gene to support multiple functional outcomes. (B-LS1-1)

LS3.A: Inheritance of Traits

Biology 1	Biology 2
- Explain that gene's nucleotide sequence is used to make proteins through transcription of DNA into mRNA and translation of mRNA into a polypeptide. - Explain how traits are inherited and expressed, using central dogma reasoning. - Explain that changes in a gene's sequence can result in changes to a protein and lead to differences in traits.	- Describe how regulatory DNA regions can affect when, where, or how much a gene is expressed. - Recognize that epigenetic influences can alter gene expression without changing the nucleotide sequence. - Explain that many traits are polygenic, leading to complex genotype–phenotype relationships. <i>Extended Knowledge</i> - Understand that similar genetic changes can have different trait effects in different cell types or environmental conditions.

B-LS3-2. Make and defend a claim based on evidence that inheritable genetic variations may result from (1) new genetic combinations through meiosis, (2) viable errors occurring during replication, and/or (3) mutations caused by environmental factors.

LS3.B: Variation of Traits

Biology 1	Biology 2
- Explain how independent assortment and crossing over during meiosis contribute to new combinations of alleles. - Explain that mutations can introduce new genetic variants that may be inherited. - Differentiate sources of mutation (replication errors and environmental factors) and explain how the location and type of change can influence trait effects. - Explain that environmental factors can influence how traits are expressed and affect observed patterns of traits in populations. - Explain how combinations of alleles can account for patterns of trait variation and distribution in populations.	- Explain how genetic factors (such as multiple loci) and environmental factors (such as resource gradients) together shape the distribution of traits in populations. (B-LS3-3) - Explain how recombination and mutation provide heritable variation, while environmental influences on trait expression (not inheritance) modulate observed traits across time and conditions. (B-LS3-3)

B-LS3-3. Apply concepts of statistics and probability to explain the variation and distribution of expressed traits in a population.

LS3.B: Variation of Traits

Biology 1	Biology 2
• Explain how independent assortment and crossing over during meiosis contribute to new combinations of alleles. • Explain that mutations can introduce new genetic variants that may be inherited. • Differentiate sources of mutation (replication errors and environmental factors) and explain how the location and type of change can influence trait effects. • Explain that environmental factors can influence how traits are expressed and affect observed patterns of traits in populations. • Explain how combinations of alleles can account for patterns of trait variation and distribution in populations.	• Explain how genetic factors (such as multiple loci) and environmental factors (such as resource gradients) together shape the distribution of traits in populations. (B-LS3-2) • Explain how recombination and mutation provide heritable variation, while environmental influences on trait expression (not inheritance) modulate observed traits across time and conditions. (B-LS3-2)

LS4 – Biological Evolution: Unity and Diversity

B-LS4-1. Communicate scientific information that common ancestry and biological evolution are supported by multiple lines of empirical evidence.

LS4.A: Evidence of Common Ancestry and Diversity

Biology 1	Biology 2
• Use multiple lines of evidence, including fossils, comparative anatomy, genetics and molecular biology, and embryology, to support claims that species share common ancestry. • Explain how similarities and differences in DNA sequences provide evidence consistent with shared ancestry among organisms. • Explain how similarities and differences in amino acid sequences provide evidence of relatedness among organisms. • Describe how anatomical and embryological features are used as evidence to support common ancestry. • Describe evolutionary relationships as branching patterns that represent lines of common ancestry.	• Explain how multiple lines of evolutionary evidence, including fossil, molecular, anatomical, and embryological data, are compared for their consistency and explanatory power. • Explain how molecular sequence data can indicate degrees of relatedness and resolve relationships among organisms. • Describe branching patterns as models of divergent evolutionary lineages, including the assumptions and limitations of these representations. • Explain how evidence from different sources can be combined to strengthen, refine, or challenge claims about common ancestry. <i>Extended Knowledge</i> • Recognize that reconstructions of evolutionary relationships are hypotheses that may be revised as new evidence emerges.

ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World

Biology 1	Biology 2
• Identify a molecular technology used to determine relatedness among organisms and note the biological question it addresses. • Compare two molecular approaches by classifying the kind of information each provides for inferring evolutionary relationships using qualitative terms only. • Explain how using molecular evidence can clarify or reorganize groups compared with classifications based only on observable traits. • Evaluate one practical implication of adopting molecularly informed classifications for society or the natural world (for example conservation targeting, invasive-species recognition, disease tracking) and state a trade-off accepted in that choice.	• Explain how advances in technology have expanded the types of evidence available to support evolutionary explanations. • Explain how new technologies can lead to changes in scientific understanding of evolutionary relationships. • Explain how changes in available evidence can lead to revisions of scientific models of evolutionary history. • Describe how the interpretation and use of evolutionary evidence can influence decisions in areas such as conservation, medicine, or biotechnology.

B-LS4-3. Apply concepts of statistics and probability to support explanations that organisms with an advantageous heritable trait tend to increase in proportion to organisms lacking this trait.

LS4.B: Natural Selection

Biology 1	Biology 2
• Explain that natural selection occurs only when variation exists among individuals within a population. • Explain that inherited differences in traits can lead to differences in survival or reproductive success under specific environmental conditions. • Explain how differential survival and reproduction among individuals provide the conditions necessary for natural selection to operate in populations.	• Explain how inherited variation and environmental pressures interact to shape changes in trait distributions across generations. • Explain how population-level patterns can be interpreted to determine whether observed changes are consistent with natural selection. • Use the Hardy–Weinberg model to explain how stable allele frequencies reflect equilibrium conditions and how deviations from expected frequencies can indicate evolutionary influences such as natural selection. <i>Extended Knowledge</i> • Explain why natural selection may be weak, absent, or difficult to detect in some populations. • Recognize that selection can act simultaneously on multiple traits with competing outcomes.

LS4.C: Adaptation

Biology 1	Biology 2
• Explain that adaptation results from natural selection acting on inherited variation within a population. • Explain that adapted populations are characterized by traits that increase survival and reproduction (fitness) in a specific environment. • Describe adaptation as a population-level outcome that emerges over many generations rather than a change within individuals. • Explain how environmental change can affect population success depending on which traits are present. • Describe adaptation, extinction, and divergence as alternative long-term outcomes of population responses to environmental change.	• Explain adaptation as a population-level outcome resulting from differential survival and reproduction under specific environmental conditions. • Explain how changing or conflicting selective pressures can influence adaptive outcomes in populations. • Explain why adaptation does not guarantee population persistence when environmental change is rapid or severe. (B-LS4-6) <i>Extended Knowledge</i> • Understand that limits on genetic variation, population size, or time can constrain adaptive potential. • Recognize tradeoffs among traits that can limit adaptive success across environments.

B-LS4-6. Create or revise a simulation to test a solution to mitigate adverse impacts of human activity on biodiversity.

LS4.C: Adaptation

Biology 1	Biology 2
Describe adaptation, extinction, and divergence as alternative long-term outcomes of population responses to environmental change.	Explain why adaptation does not guarantee population persistence when environmental change is rapid or severe. (B-LS4-3)

LS4.D: Biodiversity and Humans

Biology 1	Biology 2
- Explain that humans depend on biodiversity for resources and benefits provided by living systems. - Describe how human activities such as habitat destruction, overexploitation, pollution, invasive species introduction, and long-term changes in climate patterns can reduce biodiversity. - Explain how reductions in biodiversity can disrupt ecosystem functioning and productivity, affecting humans and other organisms. - Explain why sustaining biodiversity supports continued ecosystem functioning. - Describe tradeoffs between short-term human benefits and long-term impacts on biodiversity and ecosystem functioning.	- Explain how evolutionary processes such as adaptation, speciation, and extinction contribute to changes in biodiversity over time. - Explain the consequences of biodiversity loss for ecosystem resilience and stability. <i>Extended Knowledge</i> - Explain how cumulative human impacts across systems and scales can intensify biodiversity loss. - Recognize that biodiversity recovery depends on the magnitude of disturbance and ecological context.

ETS1.B: Developing Possible Solutions

Biology 1	Biology 2
- Identify criteria for reducing a biodiversity impact and list the constraints (for example, cost, safety, reliability, and aesthetics, and consider social, cultural, and environmental impacts, etc.). - Compare feasible options by classifying how each meets the prioritized criteria and respects constraints, including social, cultural, and environmental effects. - Identify the single most limiting constraint for this problem and explain how it should inform the next revision choice. - Explain which option best fits the criteria using evidence from biodiversity outcomes. - Analyze how one targeted refinement would improve outcomes and name the trade-off accepted to keep within constraints.	- Identify criteria and constraints (for example, cost, safety, reliability, and aesthetics, and to consider social, cultural, and environmental impacts, etc.) for the biodiversity problem in clear, measurable terms. - Classify proposed solutions by the features that align with prioritized criteria and by anticipated social, cultural, and environmental effects. - Explain how insight from a physical model or simple computer output supports keeping or dropping one design feature. - Analyze how a small refinement that improves fit to the criteria while acknowledging one trade-off (for example cost, reliability, access). <i>Extended Knowledge</i> - Recognize when a simulation result is sensitive to a single input and state a safe qualitative range for that input. - Identify a case where a physical model insight changes which simulation option should be favored and state the reason.

ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World

Seventh Grade and Biology 1	Biology 2
Seventh Grade - Sort examples of human activity by their short- and long-term consequences, positive or negative, for people and for the natural environment. - Explain how scientific understanding of matter, energy, or environmental processes can change which technologies are adopted or phased out. - Evaluate how two mitigation technologies differ in effectiveness based on regional resources and societal priorities. Biology 1 - Identify a molecular technology used to determine relatedness among organisms and note the biological question it addresses. - Evaluate one practical implication of adopting molecularly informed classifications for society or the natural world (for example conservation targeting, invasive-species recognition, disease tracking) and state a trade-off accepted in that choice.	- Identify how interconnected technological systems (for example wildlife corridors, remediation, diagnostics, precision-agriculture) depend on biological knowledge to increase benefits and reduce risks. - Classify system features by their expected benefits, risks, and constraints for the stated population or ecosystem. - Explain which system design better aligns with goals for biodiversity and stability using qualitative reasoning. - Appraise a small system modification that would improve fit to community needs while noting a trade-off. - Recognize when a different priority would change the preferred system and state the reason.